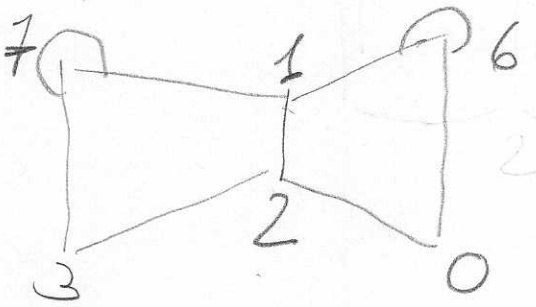
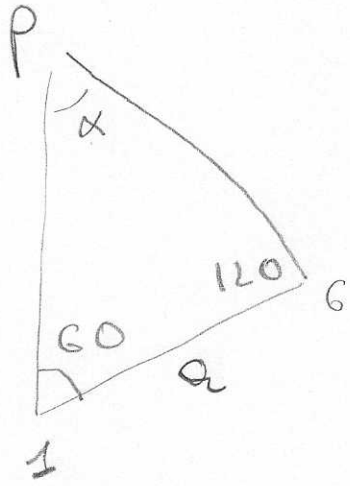
Exclusive rate II

A subset of the net is



Moreover there is a 3rd concave angle.
So the pin P has to satisfy
the condition that the angle
is larger than $\frac{2\pi}{3}$ and that
its antipodal is not contained
in 7321 ∪ 1206

For the first condition:



$$\cos\alpha = \sin 60 \sin 120 \cos a - \cos 60 \cos 120$$

$$\cos\alpha = \frac{3}{4}\cos a + \frac{1}{4}$$

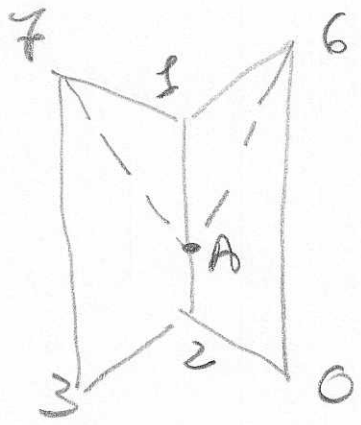
$$120 \geq \alpha \geq 60 \Rightarrow -\frac{1}{2} \leq \cos\alpha \leq \frac{1}{2}$$

$$\Rightarrow -1 \leq \cos a \leq \frac{1}{3}$$

Note that $\cos a$ corresponds to a being the side of the square

So this means that 16 is smaller than 60 in the quadrilateral 1602

Now the second pin P is at the intersection of the bisectors of the angles 6 and 7



Because 16 is smaller than 06, the bisector of 6 exits the quadrilateral in a point A on the longer edge 12

By symmetry A belongs also to the bisector of 7. So A is the new pin